Automatic Question-Answer Generation for Education on Indonesian Texts : A Review of Methodologies, Dataset and Evaluation Metrics

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***Abstract*— The advancement of research in the Automatic Questions and Answers Generation (AQAG) is notable, with significant strides being made. Diverse methodologies have been employed across different languages, each showcasing distinct traits that necessitate methodological adjustments for effective question answering. The systematic literature review (SLR) that has been developed may serve as a valuable asset for informing future research endeavors focused on formulating inquiries within Indonesian textual contexts. The literature search process, conducted through platforms such as IEEE Xplore, Springer Link, and Science Direct, yielded 63 relevant studies using keywords like *Indonesian text question generation*, and *question and/or answer generation*. The inclusion criteria applied a search from the years 2022 to 2024, while the exclusion criteria filtered out AQAG-related articles using text datasets and reputable journals, as well as studies in languages other than Indonesian. Based on the search and the application of exclusion and inclusion criteria, 46 relevant review papers were identified. The analysis results from the study indicate that AQAG (Automatic Question and Answer Generation) for education using Indonesian texts still has several research gaps that need to be addressed. One of these gaps is the limited availability of open datasets related to the educational field, especially in the Indonesian language. Additionally, another shortcoming of AQAG is that the quality of questions in certain domains is still not entirely relevant to the intended domain, resulting in questions that tend to emphasize syntactic aspects rather than accurate information extraction. Future research could combine several deep learning models to create questions and answers that are appropriate for the targeted domain.**

Keywords*— *indonesian text generation, question answer generation, systematic literature review.

I. INTRODUCTION

Manually creating or coming up with questions is a time-consuming task. It can be exhausting for question creators as they have to consider a wide range of variations based on the available material or information. Consequently, the automation of question generation, known as AQAG, has become a stimulating topic among Natural Language Processing (NLP) researchers [1]. Automating the process of generating questions and answers can significantly enhance work efficiency, especially in the field of education [2]. The primary objective of an AQG system is to create questions that are both syntactically and semantically accurate and relevant to the specific use case context. For instance, in certain scenarios, the aim is to produce questions related to specific topics of interest or derived from various segments of text within a passage [3]. Conversely, in conversational systems like chatbots that generate questions, it is essential to align with the conversation's context and keep the user engaged at the same time [4] For example, a chatbot used as an educational medium about stunting, which is a global public health issue [5]. Utilizing AQAG can enhance the assessment process in Massive Open Online Courses (MOOCs) by creating questions from the provided course material, thus making the process more interactive, adaptable, and efficient[6].

Many techniques have been utilized in the past for automating question generation, but not all of them are universally applicable across different languages and contexts. Hence, there is an ongoing debate regarding the scope of research in this area, especially as it pertains to specific languages or fields [7]. To address this, a Systematic Literature Review (SLR) will be conducted to explore the development of automated question generation in the realm of education in Indonesia. The primary goal of this SLR is to assess the strengths and weaknesses of previous research methodologies, as well as to pinpoint areas where further investigation is needed in order to enhance the overall quality of research.

Study on SLR regarding the topic of generating questions in the Indonesian language has been previously conducted by several researchers. Although there have been existing studies on this topic, there is still a need for more specific discussions focused on generating questions in Indonesian. Building upon previous SLRs with related themes, further research in this area is warranted.

An SLR using Deep Learning techniques for automatic text summarization has been conducted. The discussed journals used datasets in English, comparing the methods used

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in each journal and briefly explaining the existing methods, leading to the conclusion that applying Deep Learning techniques can produce better abstract text summaries [8]. Another study, titled "A Review on NLP-based Automatic Question Generation (AQG)," aimed to generate all possible questions from deep or high-quality text datasets. This research discusses issues, challenges, and comparative methods for creating automatic questions[1]. Furthermore, there is a study titled "AQG: A Review of Methodology, Dataset, and Evaluation Metrics" [9]. Three scenarios considered are generating questions independently, from visual stimuli, and in conversational settings. Some suggestions for improvements so that can produce more questions experience that has associated significance with the document or text the original. as well as the necessity of combining models to produce results method making the question more automatic good.

II. Research Methods

Three phases were executed to prepare for this SLR. The first stage involves planning the literature review, conducting the review, and then presenting the review results.

A. Planning Literature Review

The stages of planning a literature review contain information about determination objectives and research questions. Then, the determination of search strategy, criteria inclusion and exclusion. Details of each stage are explained in points B to D.

B. Determining Goals and Research Questions

The prevalence of study-related AQAGs is extensive. Each language exhibits unique characteristics, necessitating adjustments to the generation method based on the language in use. It is imperative to conduct development research on question generation, particularly within the Indonesian context. Therefore, conducting a SLR to delve into question generation aspects, particularly tailored for the Indonesian populace, is crucial. Research questions are formulated to address the objectives of this SLR, covering four inquiries spanning from Research Question 1 (RQ1) to Research Question 4 (RQ4). RQ1: What are possible methods used for AQAG?. RQ2: What is the only available dataset for the Indonesian AQAG language?. RQ3: How do evaluation results make questions and answers automatic?. RQ4: What are AQAG 's challenges and newness in general or AQAG in the text language Indonesia?.

1) Search Strategy

Search studies about generation inquiries are conducted via IEEE Xplore, Springer Link, and Science Direct. Keywords used are "question generation," "question answer generation," " Indonesian text question generation," and "automatic question generation." The four keywords used alternate for maximizing search study related to AQAG.

2) Inclusion and exclusion criteria

Criteria inclusion is a criterion used in condition-analyzed studies. Criteria inclusion on this SLR is a study about question-answer generation and research published in the range time between 2022 to by 2024. Meanwhile, criteria exclusions used are studies about question-answering generation with text datasets and journals nor conference reputation and language besides Indonesian.

C. Literature Review Proces

The data search using the inclusion selection method yielded 63 documents. After further selection using exclusion criteria, 17 studies were found to be irrelevant. This included the removal of 5 non-reputable articles and an additional 12 articles based on the question generation research with text datasets, leaving 46 documents. Of these, there were 20 studies from 2022, 21 studies from 2023, and 5 studies from 2024.

D. Reporting the Results of the Literature Review

The review findings are presented alongside a detailed analysis of the primary study's overview and response to each research question. Each research question's description is derived from the results obtained through data extraction.

III. Results And Discussion

A. Automatic Question-Answer Generation Overview

The role of AQAG is pivotal in the education sector, as indicated by the findings outlined in Table 1 of the literature review, it is projected that between 2022 and mid-2024, researchers will conduct 21 out of the 46 studies examining AQAG in education domain. This paper focuses on discussing how AQAG can enhance the education domain through its automation processes, supporting question generation, evaluation, learning, and improving understanding.

TABLE I. Domains in Question Generation

Domain	Description	References
Geology	AQAG in geology is used for information about studying questions about Earth's plane, like the environment's place of mineral deposits.	[10]
Health	In the health domain, AQAG can guide the diagnostic and upgrade process utilization source Power Medical with reduced dependency on engagement doctors to get information about health.	[11], [12], [13]
Open Domain	Open domain in AQAG refers to fields or contexts that are not limited by topic certain, possible discussion in various fields like science, news, culture or popular discussion.	[14], [15], [16], [17], [18], [19], [20], [21], [22], [23]
Education	AQAG in the education domain is related to the learning and teaching process. This domain includes various disciplines of knowledge education at various levels, starting from education base until education height and training professional.	[17], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43]
Grammar	AQAG in domain grammar aids in understanding and forming correct and meaningful sentences by teaching the governing structures and rules of language use in communication, encompassing aspects such as morphology, syntax, semantics, and pragmatics.	[44], [45], [46]

The use of AQAG in education is broad and impactful. It can create learning chatbots that adapt to students' needs, offering simpler questions or additional examples when concepts are challenging. These chatbots adjust question difficulty, provide tailored feedback, and enhance understanding. AQAG also automates practice exercises and exams by generating questions from uploaded teaching materials, covering various difficulty levels and saving teachers time. Moreover, AQAG supports project-based learning by generating questions that promote critical thinking and in-depth exploration of topics.

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B. *Classification of Automatic Question-Answer Generation Techniques*

Automated systems use various methods to generate questions and answers, influenced by the types of questions and the difficulty of generating answers. Rule-based systems employ specific NLP techniques, while neural network-based models use precise representations during training. Based on the data in Table 2, the type of question formed is known, namely, yes/no type question, factoid, non-factoid, and multiple-choice.

TABLE II. TYPES OF QUESTIONS IN THE QUESTION GENERATOR

Question Type	Description	References
Factoids	Question type: This direct results answer describes facts or information answered in a short and precise way. Examples are factoid questions like the prefixes "who", "when" and "how much".	[13], [14], [16], [19], [21], [22], [23], [28], [31], [32], [45], [47], [48], [49], [50], [51]
Non factoid	Non-factoid questions cannot be answered with short answers like factoid questions. They need analysis, opinion, or an answer with a deep explanation. For example, non-factoid " why " or " how " questions.	[13], [25], [27], [28], [30], [34], [35], [36], [40], [43], [46], [47], [49], [52], [53]
Yes/No Type Question	This type of essential inquiry demands a binary response, either yes or no. Such questions necessitate deeper contemplation to ensure accurate and reasoned responses.	[43], [47], [54]
Multiple Choice	This type will give a number of answer options. From several options, only one will be true. Question choice is designed to test knowledge or understanding in an effective and efficient way.	[37], [38], [42], [43], [47]

Based on the analysis of 46 studies in the field of AQAG, it can be broadly concluded that there are three distinct characteristics of the models that have been applied. Each of these models has different strengths and techniques for generating questions and answers, as shown in Table 3. In addition to these three models, combinations of several models and techniques are often used to achieve the best results.

TABLE III. METHODS THAT HAVE BEEN APPLIED TO AUTOMATIC QUESTION-ANSWER GENERATION

Method	Description	References
Encoder-Decoder	The Encoder-Decoder model typically employs supervised learning to improve the machine's ability to generate relevant responses to questions. This involves training the model with substantial data consisting of paired questions and answers, allowing it to understand queries and produce accurate responses. The encoder's function is to transform the question into dense representation vectors, while the decoder generates responses based on the representation vectors provided by the encoder.	[12], [16], [20], [30], [31], [34], [35], [38], [39], [45], [48], [50], [53]
Deep Reinforcement Learning (DRL)	Application of DRL to generate questions involves making an agent or models that can build relevant and informative questions based on context or information provided as well as a possible system to produce more answers contextual and informative in an automatic.	[22]
Transformers	Transformer models have beenone leading approach in AQAG-based language. This approach uses robust Transformer architecture for processing and understanding text and producing answers based on given questions. Its abilities to catch information in complex contextual contexts and modelling connections between words in text are impressive. The transformer models used in AQAG include BERT, T5, RoBERTa, and GPT-3.	[10], [11], [12], [14], [15], [18], [19], [21], [22], [23], [25], [26], [27], [28], [29], [32], [33], [35], [36], [37], [40], [41], [42], [43], [44], [46], [47], [49], [51], [52], [55]

The popular model used today is the Transformer model. As shown in Fig.1, the Transformer model is widely used across various domains. In the education domain, four journals utilize Indonesian text [18], [23], [32], [35].

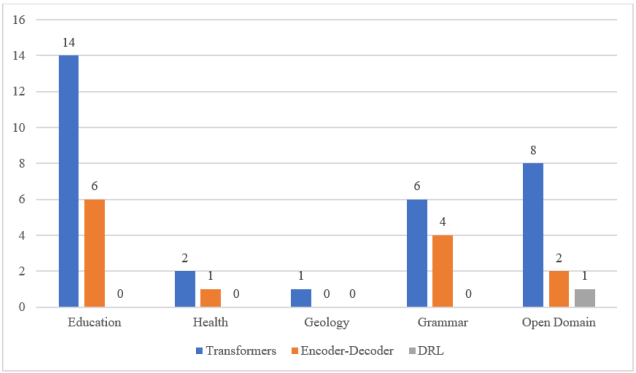


Fig. 1. Most Popular Methods in Various Domains

C. *Datasets*

Many data sets have been opened to support studies in making questions and answers automatically, but for open data in the education domain, Indonesian data still needs to be made more available. Table 4 is the dataset for making question-and-answer Indonesian in the education domain.

TABLE IV. SUMMARY OF DATA SETS FOR AQAG

References	Dataset Name	Detail Dataset
[32]	SQuAD, TyDiQA, and IDK-MRC, which stand for Stanford Question Answering Dataset, Typologically Diverse Question Answering, and Indonesian Machine Reading Comprehension respectively.	The SQuAD V2.0 was translated into Indonesian using the Google Translate API V2, resulting in 536 articles and 161,550 question-answer pairs. Additionally, 200,000 pairs of questions and answers were obtained from TyDiQA.
[35]	SQuAD, TyDiQA	SQuAD has 10,567 test data entries, while TyDiQA has 550 test data entries.
[18]	TyDiQA	Language data extraction Indonesia from TyDiQA Dataset.
[23]	AC-IQuAD stands for Automatically Constructed Indonesian Question Answering Dataset.	AC-IQuAD was created using the Indonesian Wikipedia Corpus, comprising 134,000 simple questions and 60,000 complex questions.

D. *Evaluation Techniques for Question-Answer Quality*

Questions and Answers (QA) generated by the model must be evaluated to determine the quality of QA so that the QA produced makes sense for the intended purpose or domain. Hence, evaluation methods generally fall into two categories: automated assessment and expert-driven or human-centered evaluation. Some studies incorporate a combination of both

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evaluation approaches. The most commonly used evaluation methods are evaluation metrics or automatic evaluation using methods such as BLEU, ROUGE-L, METEOR, and F1-Score, and of course, there is also expert assessment. However, the expert-based evaluation itself cannot be standardized because each expert has a different level of understanding. Table 5 shows the evaluation methods used by each study.

TABLE V. EVALUATION ON QUESTION ANSWER GENERATION

Technique Evaluation	Description	References
Automatic evaluation	AQAG serves as a significant milestone in assessing model performance without constant human intervention. Various metrics and methodologies, including BLEU, ROUGE-L, Meteor, and F1-Score, are employed in AQAG evaluation. The utilization of metrics focusing on semantic similarity or gauging the relevance of responses can enhance the accuracy and relevance of the evaluation process.	[10], [11], [12], [13], [14], [15], [16], [18], [19], [20], [21], [22], [25], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [39], [40], [41], [42], [44], [45], [46], [47], [49], [50], [51], [52], [53], [54], [55], [56]
Human-based evaluation	An expert evaluator examines the quality of questions and answers produced by the model, offering insights into aspects that automatic metrics might overlook, such as contextual appropriateness, clarity, and semantic relevance. Human-based evaluation is crucial for comprehensively assessing the performance of AQAG models.	[12], [13], [20], [21], [23], [26], [27], [34], [38], [39], [41], [42], [43], [47], [48], [51], [54], [56], [57]

Evaluation metrics typically refer to methods for measuring how effectively a generation system produces relevant, informative, and contextually appropriate questions and answers. In automatic evaluation, two primary metrics are used: precision and recall. Precision assesses specificity, while recall measures sensitivity. These metrics are commonly employed in popular automatic evaluation methodologies. Figure 2 illustrates the review results of this study on the popular metrics used in the last three years for evaluating AQAG..

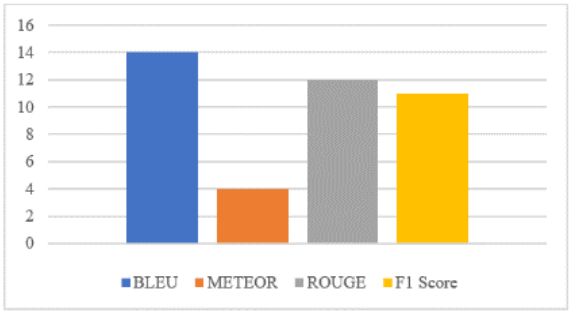


Fig. 2. Automatic Evaluation Metrics Used in Research Over the Last Three Years

In addition to those metrics, there are also evaluation metrics based on Transformer model approaches such as BERT, which is used as both AQAG and automatic evaluation models [11], [25].

E. Challenges and Future Directions

- Challenges to AQAG

Combining deep learning models with processing language experience has made great progress in automating the creation of questions and answers. However, several challenges and possibilities for innovation in the future in AQAG are explained in Table 6.

TABLE VI. CHALLENGES ON AQAG AND ON AQAG WITH INDONESIAN SUBTITLES

Draft	Description	References
Quality of Questions	Producing questions with proper syntax has been aided by using language models in related contexts. However, thorough investigations have shown that, beyond simply generating questions, a sufficient number of inquiries about semantics and relevance are also necessary. Additionally, some extensive methodologies emphasize syntax over extracting meaningful information from questions, which complicates generating meaningful questions. While syntax is important, it is essential to ensure that the questions generated can extract relevant information. Therefore, comprehensive checks are required.	[11], [12], [13], [16], [18], [22], [25], [32], [36], [41], [45], [49], [50], [52], [53], [54], [55], [57]
Datasets challenge	At the moment, available datasets are in English. The source language for language datasets in Indonesia is still English, which is then translated into Indonesian. Several datasets focus on specific domains, but they do not publish or provide an open dataset, so they do not yet have a dataset that is tested on all models.	[10], [14], [18], [20], [22], [23], [27], [29], [40], [41], [46], [53]
Metrics challenge	Standard metrics like BLEU, METEOR, ROUGE, and F1-Score are often used to evaluate generated questions automatically. However, there is a need for more comprehensive metrics that include factors like the user's language proficiency and the criteria for assessing the syntax and grammar of QA pairs. Although many studies use expert evaluation, incorporating more languages is essential for better context understanding. Developing efficient metrics that can automatically estimate the quality of generated questions is crucial to reduce reliance on expert evaluators.	[13], [21], [28], [29], [30], [31], [33], [34], [38], [42], [44], [46], [48], [50], [55], [58]

- Future Research Direction

Most available datasets have an open domain, like Wikipedia, a social networking platform, and others. As a result, several recent studies focus on the transfer of learning from one domain to another. For example, in case [14], [15], transfer training is carried out using non-educational datasets such as Natural Question (NQ) and SQuAD for objective answer questions in the education domain. This shows that transfer learning is beneficial in situations where data for training is not enough. Using architecture pre-workout, such as BERT or GPT-based studies, can identify the difference between AQAG in education and non-education. Therefore, that would be useful for using transfer learning on AQAG in field education.

Because in this part, a large dataset for AQAG is derived from open datasets such as Wikipedia or SQuAD. So, that dataset should not be suitable for use in domain-based AQAG. Specialized domains such as education and medical fields have additional requirements beyond the sole focus on generating questions relevant to manufacturing. For instance, within the education sector, there is a necessity to generate questions that correspond to a specific cognitive level or align with a particular category. Conversely, within the medical domain, questions must be tailored to address accurate diagnosis. As one of the for [59] example, researchers collaborated with 10 medical experts to compile a dataset containing over 2000 questions relevant to diagnosis.

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Following the analysis of question types, they employed pre-trained transformer models to train on the dataset, yielding promising outcomes. Therefore, the quality domain-based corpora must be made with an approach according to the domain raised.

IV. Conclusion

This study centers on the implementation of a SLR within the realm of question generation, specifically focusing on the Indonesian language. A total of 46 studies compliant with the inclusion and exclusion criteria and demonstrating high quality for analysis were identified. Within AQAG, three methodological characteristics are prevalent: Encoder-Decoder, Deep Reinforcement Learning, and Transformers. While each model has its merits, Transformers emerge as the most widely utilized model, delivering automatic question and answer generation with superior accuracy results. A substantial research gap exists within the AQAG domain, particularly concerning the scarcity of open datasets in the Indonesian language designed for educational applications. Additionally, the relevance and quality of generated questions often prove insufficient, as they tend to prioritize syntactic accuracy over extracting meaningful information. Furthermore, the evaluation metrics frequently fall short in assessing the quality of the generated questions. Therefore, human-based evaluation remains crucial to assess context, clarity, and relevance in all domains.

Future research should focus on developing more robust models that incorporate various techniques, creating high-quality domain-specific corpora, and enhancing evaluation metrics to reduce reliance on human assessment. Additionally, there is a need to explore transfer learning to adapt models to specific domains.

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